

Exp 21

Q1

```
energy <- read.csv("C:\\Users\\91637\\OneDrive\\Documents\\Exp 21.csv")
```

```
hist(energy$Units_Consumed,  
     col="skyblue",  
     main="Histogram of Units Consumed",  
     xlab="Units Consumed")
```

```
plot(density(energy$Units_Consumed),  
     col="red",  
     lwd=3,  
     main="Density Plot of Units Consumed",  
     xlab="Units Consumed")
```

#Q2

```
library(ggplot2)
```

```
energy <- read.csv("C:\\Users\\91637\\OneDrive\\Documents\\Exp 21.csv")
```

```
ggplot(energy,  
       aes(x=Temperature,  
           y=Units_Consumed,  
           size=Peak_Hours)) +  
geom_point(alpha=0.6,  
           color="blue") +  
labs(title="Temperature vs Units Consumed",  
     x="Temperature (°C)",  
     y="Units Consumed (kWh)") +  
theme_minimal()
```

```
library(dplyr)  
library(ggplot2)
```

```
energy <- read.csv("C:\\Users\\91637\\OneDrive\\Documents\\Exp 21.csv")
```

```
avg_renewable <- energy %>%  
  group_by(Sector) %>%  
  summarise(Average_Renewable = mean(Renewable_Usage))
```

```
print(avg_renewable)
```

```
ggplot(avg_renewable,  
       aes(x=Sector,  
           y=Average_Renewable,  
           fill=Sector)) +  
geom_bar(stat="identity") +  
labs(title="Average Renewable Usage by Sector",  
     x="Sector",
```

```
y="Renewable Usage (%)" +  
theme_minimal()
```



